

PhaseQ Daily Schedule

Phase Q — Day-by-Day Schedule · Quantum Computing + Quantum Machine Learning

Weeks 76-86 · Mon Feb 14 → Sun May 1, 2028 · ~335 hrs · off the job-critical path (interview in parallel)

Goal: match your hands-on VQE/ADAPT/DMET work with rigorous theory — derive VQE from the variational principle, explain barren plateaus, defend ansatz/measurement choices, and give an honest QML take (Schuld). Also grounds Phase-5D physics.

COI: public molecules only (H₂/LiH/H₂O or a QM9 subset). No Lilly molecules (no keto-enol/HATU specifics), nothing from the vendor engagement.

*Blocks: **A** 06-08 (theory, derive) · **B** 08:30-10:30 (PennyLane/Qiskit coding) · **Evening** 20:30-22:30 (read, Anki). Weekend = buffer + rest. Threads light: **T/R** kept minimal — **you're interviewing in parallel**, so protect energy; pull Phase Q forward/interleave if a QML role appears earlier.*

Week 76 · Feb 14-20 — Intuition entry + group theory / symmetry

Day	Block A	Block B	Evening
Mon	Quantum Country (spaced primer); MIT 8.04 intro	Qubit states in code (numpy)	Anki
Tue	Complex vector spaces; bra-ket	Bloch-sphere visualization	Anki
Wed	Groups & representations	Symmetry operations in code	Anki
Thu	Lie groups; link to equivariance (GDL)	Rotation reps	R: light applications
Fri	Why symmetry underlies operators & equivariant nets	Review	Whiteboard-Fri: groups/representations
Sat-Sun	Buffer + rest		

Week 77 · Feb 21-27 — Quantum-mechanics math

Day	Block A	Block B	Evening
Mon	Hilbert spaces; inner products	State vectors in code	Anki
Tue	Operators; Hermitian/unitary	Operator algebra	Anki
Wed	Tensor products; multi-qubit states	Tensor two qubits	Anki
Thu	Measurement; eigenvalues as outcomes	Measurement simulation	R: applications
Fri	Pure vs mixed; density matrix	Density-matrix code	Whiteboard-Fri: tensor two qubit states; unitarity
Sat-Sun	Buffer + rest		

Week 78 · Feb 28-Mar 5 — QC fundamentals: Basics of Quantum Information (Watrous/IBM)

Day	Block A	Block B	Evening
Mon	Single-qubit gates	Qiskit: single-qubit circuits	Anki
Tue	Multi-qubit gates; entanglement	Bell states	Anki
Wed	Circuit model	Build/measure circuits	Anki
Thu	Superdense coding / teleportation	Implement teleportation	R: applications
Fri	Review	Circuit drills	Whiteboard-Fri: entanglement; a 2-qubit gate
Sat-Sun	Buffer + rest		

Week 79 · Mar 6-12 — Fundamentals of Quantum Algorithms

Day	Block A	Block B	Evening
Mon	Deutsch-Jozsa; oracles	Qiskit: DJ	Anki
Tue	Quantum Fourier Transform	Implement QFT	Anki
Wed	Phase estimation	Implement QPE	Anki

Thu	Grover (concept)	Grover demo	R: applications
Fri	What each algorithm teaches	Review	Whiteboard-Fri: QPE walkthrough
Sat-Sun	Buffer + rest		

Week 80 · Mar 13-19 — General formulation + error-correction intro

Day	Block A	Block B	Evening
Mon	Density matrices; quantum channels	Channel simulation	Anki
Tue	POVMs; noise models	Noise in Qiskit	Anki
Wed	Error-correction basics (bit/phase flip)	3-qubit code demo	Anki
Thu	Error mitigation (near-term)	Mitigation demo	R: applications
Fri	Why noise matters for VQE	Review	Whiteboard-Fri: a channel + POVM
Sat-Sun	Buffer + rest		

Week 81 · Mar 20-26 — Variational algorithms: VQE / ADAPT-VQE

Day	Block A	Block B (PennyLane)	Evening
Mon	Variational principle (derive)	VQE on H ₂ (UCCSD)	Anki
Tue	Ansätze; parameter-shift rule	Parameter-shift gradients	Anki
Wed	ADAPT-VQE (operator pool)	ADAPT-VQE on LiH	Anki
Thu	Optimizers for VQE	Compare optimizers	R: applications
Fri	Why ADAPT is parameter-efficient	Review	Whiteboard-Fri: derive VQE from the variational principle
Sat-Sun	Buffer + rest		

Week 82 · Mar 27-Apr 2 — Variational: DMET-VQE + QAOA + measurement

Day	Block A	Block B	Evening
Mon	DMET fragmentation	DMET-VQE on a small system	Anki
Tue	Classical-quantum interface	Fragment + embed	Anki
Wed	QAOA (combinatorial)	QAOA on Max-Cut	Anki
Thu	Measurement: Pauli grouping, shot budget	Grouping demo	R: applications
Fri	When DMET is worth the overhead	Review	Whiteboard-Fri: the classical-quantum interface
Sat-Sun	Buffer + rest		

Week 83 · Apr 3-9 — QML theory (kernels, barren plateaus, Schuld critique)

Day	Block A	Block B	Evening
Mon	QML as kernel methods	A quantum-kernel demo	Anki
Tue	Barren plateaus (math)	Gradient-variance experiment	Anki
Wed	Mitigations (init, local cost)	Local-cost demo	Anki
Thu	Schuld "taking stock" critique	Review claims critically	R: applications
Fri	The honest hype-vs-promise take	Review	Whiteboard-Fri: a barren-plateau argument
Sat-Sun	Buffer + rest		

Week 84 · Apr 10-16 — QML theory cont + encodings

Day	Block A	Block B	Evening
Mon	Data encodings (angle/amplitude/basis)	Encoding demos	Anki

Tue	Expressibility & entangling capability	Expressibility experiment	Anki
Wed	Is quantum advantage the right goal?	Review	Anki
Thu	Trainability vs expressibility trade-off	PQC experiment	R: applications
Fri	QML limits, honestly	Review	Whiteboard-Fri: QML-as-kernels
Sat-Sun	Buffer + rest		

Week 85 · Apr 17-23 — Quantum chemistry on quantum computers

Day	Block A	Block B	Evening
Mon	Qiskit summer school: chemistry track	End-to-end small-molecule pipeline	Anki
Tue	Basis sets on QC; active spaces	Active-space selection	Anki
Wed	Sample-based/Krylov diagonalization (SQD)	SQD demo	Anki
Thu	Benchmarking vs classical CCSD	Compare to CCSD/6-31G	R: applications
Fri	Where QC chemistry stands today	Review	Whiteboard-Fri: VQE vs QAOA — when each
Sat-Sun	Buffer + rest		

Week 86 · Apr 24-May 1 — 🏠 Capstone Q + self-test

Day	Block A	Block B	Evening (+ R)
Mon	Design the capstone (public molecule)	VQE + UCCSD on H ₂ /LiH/H ₂ O	Anki
Tue	Add ADAPT-VQE + DMET-VQE	Run + benchmark vs CCSD	Log
Wed	Barren-plateau analysis	Ansatz/optimizer/measurement ablation	Anki
Thu	Write-up (4 pages)	Defend choices + noise mitigation	R: post
Fri	🏠 PHASE-Q SELF-TEST (derive VQE from variational principle; parameter-shift rule; barren-plateau argument; VQE vs QAOA; honest QML take; DMET interface)	Final polish	► QML-branch applications
Sat-Sun	Buffer + rest · <i>Deliverable:</i> Capstone Q (public molecules) · full-depth plan complete ~mid-2028		

End of Phase Q — the full plan is complete. QML career branch open; continue interviewing across all tiers.